

REVIEW EXERCISE 01

Question 01. Consider the following activity. A little boy is playing hide-and-seek with his friends in the playground. He is the seeker that attempts to locate all hiders, and he will yell "Found" every time he catches a hider.



Specify the PEAS description for the task environment of the above activity.

Performance measure: All hiders are found.

Environment: hiders, playground

Actuators: hands (to touch), leg (to move), mouth (to yell)

Sensors: eyes (to see and find), ears (to hear the noise made by hider)

Question 02. Identify the following task environment properties of the above task. Do not forget to give your explanation for every dimension.

Fully observable vs. **Partially observable**. Although the seeker does not see hiders, he/she can access any legal location in the playground to find the hiders, fully using his percepts.

Single-agent vs. **Multi-agents**. One seeker and at least one hider

Deterministic vs. **Stochastic**. The seeker can predict incorrectly the place hiders hide.

Episodic vs. **Sequential**. Locating a hider is independent from another hider. However, the goal is still to locate ALL hiders.

Question 03. Consider the following activity. A little boy is playing blind man's bluff with his friends in the playground. He is the catcher that attempts to touch any hider to switch their roles, while the hidere shout and laugh to distract the catcher.



Specify the PEAS description for the task environment of the above activity.

Performance measure: the catcher touches one hider to switch his/her role.

Environment: Friends, playground

Actuators: legs(to move), hands(to touch)

Sensors: ears (to hear the noise the hider makes), hands (to touch the objects)

Question 04. Identify the following task environment properties of the above task. Do not forget to give your explanation for every dimension.

Fully observable vs. Partially observable.

The catcher is blindfolded, he/she may be aware of hider, but he does not know their exact locations.

Single-agent vs. Multi-agents. one catcher and at least one hider

Deterministic vs. Stochastic.

the catcher cannot surely guess where the hidere are, his prediction may not correct

Episodic vs. Sequential.

The catcher just needs to catch one hider. During the time, his activities may change suddenly, not following any specific pattern.